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This is part two of lecture week one, SIT 112, Introduction to Data Science and AI. If you haven't visited part one of lecture week one, please do so as soon as possible. Part one includes essential information about SIT 112, including how we run the lectures, workshops, including how we run the lectures, workshops, as well as the details of the assessment tasks and any other information that you need for this unit. All right, so the contents we are going to cover in this lecture are as follows. We first explore some definitions of artificial intelligence, also known as AI. Then we talk about foundations of AI, the risks of using or misusing AI. We talk about how can we have a human-centered approach to AI. Then we touch on the topics of machine learning and natural language processing as two of the main areas of AI which are also used in data science. Then we talk about data science and its definition, talk about the things that need to be done before data analysis and the five phases of data analysis. Then we cover some of the basics of Python for data science and AI, and introduce you to a tool named JupyterLab, which will be used in this unit for writing and executing Python code. As you can see in this one diagram, there's a big overlap between data science and artificial intelligence. And the overlap is probably much more significant than we can see in this diagram nowadays. But this diagram just shows that there is a big overlap. In fact, one may argue that data science and AI are now merged into one field, which is kind of the focus of this unit as well.

02:32

So what is artificial intelligence? To be able to define artificial intelligence we need to understand artificial and intelligence, needless to say. Artificial here means not natural, something that is made up, is a mimic. And artificial intelligence therefore means a kind of intelligence that is not natural, it's not organic. Now let's look at the definition of intelligence. There are two ways that we can define intelligence. One is based on mimicking human behavior or human performance. So this is a kind of definition that tries to create or tries to introduce a kind of intelligence that is as close as possible to how humans perform or behave. Alternatively, intelligence is defined as doing the right thing or acting rationally, acting or thinking rationally. So this definition is more mathematical, logical, abstract as a formal definition for intelligence. But the subject matter itself also varies here. In other words, we can think about intelligence the s

subject matter itself also varies here. In other words, we can think about intelligence as thinking intelligently or acting intelligently. So, if you apply this to the two definitions that we just explored together, let's take, for example, rationality. One thing is to think rationally. another thing is to act rationally. And if you think they are the same thing, remember last time you thought something was wrong to do but you still did it. So thinking rationally does not mean, does not necessarily lead to acting rationally. And vice versa, sometimes something, some intelligent agent might act rationally while the thinking behind that rational action is not rational itself. All right, so based on the two definitions of intelligence, rational as opposed to human-like, we can have two approaches to artificial intelligence. And that is either to pursue a kind of intelligence that is human-like, it mimics human behavior. In that case, we need to use psychology, because psychology is a science, if it can be called science, which many would argue it's not. But if we can use psychology to understand human behavior, that includes their thought process and their actions, then we would be able to create, manufacture a kind of intelligence that mimics human thought process and actions. Which, as it sounds, is a very sophisticated process and one might argue that it's far more difficult to achieve compared to the rationalist approach. The rationalist approach, the other hand focuses on again doing the right thing. So there is a definition of rationalism. Based on that definition an intelligence, a kind of intelligence is built or is created that uses mathematics, engineering, statistics, control theory, economy, and any other sciences that might be needed to create an intelligence or kind of intelligence that leads to thinking slash acting rationally. So let's put everything together here we have two different approaches to defining intelligence that is based on mimicking human or creating a human-like intelligence and creating rational intelligence. And of course, each of these could be concerning the thought process or the actions. So based on this, we can come up with four possible definitions for AI. Based on this, we can come up with four possible definitions for AI. Thinking humanly, acting humanly, thinking rationally, and acting rationally.

07:51

Okay, so here we have a definition of artificial intelligence based on thinking humanly, thinking like humans. So if you want to create a kind of intelligence that mimics the thought process of humans, we need to first understand the thought process of humans. So we need to understand and study human cognition. Therefore, we need to use techniques like introspection, psychological experiments, or brain imaging. And

most commonly, a combination of all these. So we need these to understand human cognition and to be able to build cognitive models. and to be able to build cognitive models. Cognitive models then will work as the engine for a human-like intelligence. That mimics human thinking. So, therefore, the interdisciplinary field of cognitive science is needed here. That brings together computer models with experimental techniques from psychology and of course brain imaging and other techniques that might help to understand human thinking. So another definition for artificially intelligent concerns acting humanly which is associated with the turing test in other words turing test is proposed to test whether the computer is acting humanly or not in fact the test is as follows there is an interrogator in the room and there's a wall. Behind the wall there's an object which could be computer or a human. If the interrogator cannot tell the difference between the responses submitted by the computer and by the human to his questions. That means that computer is intelligent enough to act humanly. All right, that's the Turing test. In fact, Turing test is a motivational test. It's motivated the study of certain fields in AI, including natural language processing, because the computer needs to understand human language to be able to respond to the queries or the questions that are put to the computer by a hypothetical interrogator. hypothetical interrogator. Knowledge representation. A computer needs to be able to digest information. Knowledge. Model them in a way that it can understand. It can process and can act upon them. Automated reasoning, the computer needs to be able to reason on the knowledge and information and to be able to answer the questions and of course the ability to learn. Machine learning, the computer needs to be able to learn from its mistakes to improve the answers that it's submitting, responses that it's submitting to get as close as possible to how a human would respond to a question from the interrogator. Some, however, have argued that this is not enough to examine human-like intelligence in terms of acting humanly. And therefore, they have proposed a new test, which is referred to as total Turing test. This also requires interaction with the objects in the real world. And that has also motivated the study of computer vision, speech recognition, and computer robotics.

12:37

Here's another definition of artificial intelligence based on thinking rationally, also referred to as laws of thought or laws of thought approach. Thinking rationally attempts to codify right thinking, that is, irrefutable reasoning processes. But what do we mean by laws of thought? In fact, we're talking about logic. We're talking about patterns fo

r argument structures that always yield correct conclusions when given correct premises. For instance, Socrates is a man and all men are mortal. The conclusion would be Socrates is mortal. This conclusion would always be true. So logic is used to create intelligence systems. And the problem or the limitation of this approach is that it requires some certainty from real world, which in many cases is absent. In fact, in most real world problems, In fact, most real world problems we don't have uncertainty about different events or concepts. So to deal with this or to somehow mitigate this limitation we use probability to allow for reasoning with uncertainty. Okay, last but certainly not least, acting rationally is another definition for artificial intelligence or is another approach to defining artificial intelligence, approach to defining artificial intelligence, which concerns rational agent. What do we mean by that? Remember from laws of thought or thinking rationally that the emphasis in laws of thought or thinking rationally is on the logic or the inference process on making correct inferences. But making correct inferences can lead to acting rationally. However, sometimes the agent or the computer or the object can act rationally without thinking rationally. That could be a reflex. For example, to recall from a hot stove. That's a correct action. That's a rational action in the potential absence of rational thinking. rational thinking. So to act rationally does not imply thinking rationally. In other words, acting rationally doesn't come with necessarily thinking rationally. The focus of a rational agent, therefore, would be on acting rationally with or without thinking rationally. The rational agent uses its senses to receive the information from the environment, process them, and act based on them. So the expectation here is a rational action or set of rational actions. Now let's talk about the foundations of AI. As we discussed earlier artificial intelligence is an interdisciplinary domain. It is based on several different areas of science and technology.

16:26

One of the main foundations of AI is philosophy. Why is philosophy important? It raises questions about a lot of the concepts that form AI and direct AI. and direct AI. For example, the very concept of artificial, what's artificial, what's natural, what's the difference between them, what is knowledge, what is thinking, what are human thoughts and how do they relate to their behavior and many other concepts that we mentioned some of them when we were defining or when we were proposing some definitions for AI. So asking these questions are important because they form the way we look at AI and the way we develop AI. Here are some examples of the kind of questions that

can be discussed from a philosophical perspective. For example, we can ask, can formal rules be used to draw valid conclusions? That concerns rationality. Does the mind arise from a physical brain? If so, how? In other words, is mind and, or are mind and rational brain the same thing? Or they are different? If they are the same thing, or if mind is part of the physical brain, then how do they relate? And the next question that we can ask is, where does knowledge come from? What's the source of knowledge? Mathematics is obviously another important foundation of AI. Mathematics especially concerns the rationality of AI. Mathematics especially concerns the rationality in AI. So the rational approach, remember from different approaches that we could have to AI. We need mathematics, we need logic for the rational approach to AI. And the questions such as what are the formal rules to draw valid conclusions? What can be computed? How do we reason with uncertain information? The concept of probability. And it's interesting that a lot of the concepts in mathematics also can be linked to philosophy. For example, the concept of correlation and causation is an interesting philosophical discussion. AI is also founded on economics, but not necessarily in its monetary sense. In fact, the economics of AI may include a broader range of human values and preferences. Economics of AI may include a broader range of human values and preferences. So we can think about questions like how should we make decisions in accordance with our preferences and values? How should we do this when these values or preferences collide or come into conflict? And how should we do this when the payoff may be far in the future? Neuroscience is another important foundation of AI, especially if you want to create artificial intelligence that can mimic human thought process, in other words thinking humanly. We mentioned that we can utilize psychology and brain imaging. Neuroscience is a way to understand how human brain works and to understand how humans think. This can be used in combination with insights from psychology to understand human thought process, to understand human cognition. to understand human thought process, to understand human cognition. It can also be used to prevent from the potential adverse impacts of AI, such as how, for instance, computers can affect human brain. It's neuroplasticity or lead to conditioning, lead to reputation compulsion and other potential dangers. And of course, there are several other areas that can be considered as foundations of AI. They include psychology, as we mentioned earlier. You can see the overlap between psychology, neuroscience and philosophy. Linguistics, how does language relate to thoughts? Computer engineering, how to build efficient computers so they could, for example, execute algorithms that can make inferences so that they can help the a

gent or the object to act rationally. How to create computers that are able to process complicated or sophisticated knowledge models. Control theory and cybernetics are also important. Questions such as how can our artifacts operate under their own control can be discussed in this regard.

21:58

So we talked about different definitions of AI, foundations of AI, and we want to acknowledge that there are risks associated with the use of AI or with the misuse of AI. And these risks could be inadvertent or otherwise. What are risks? We're going to get back to this topic later in more detail, more depth. It's important and interesting area of research and study. But at this stage we just want to acknowledge the risk and acknowledge that AI needs to be used responsibly or else there would be harmful impacts on people in society including loss of reputation, loss of money or even loss of life. So the impacts could be profound. Some examples are listed here such as risk of lethal autonomous weapons or mass surveillance or bias in decision-making. But the examples are numerous and in many cases bias in decision making. But the examples are numerous and in many cases that they find their ways to news headlines. So a human-centered approach to AI is needed to ensure a responsible AI that where the negative impacts of AI can be mitigated on people and society. Therefore human values. This is from a prominent sociopsychologist, Schwartz. And it includes 10 main categories of human values. Under each category, you can see some of the manifestations of these values. For example, privacy is a manifestation of self-direction or excitement in life is a manifestation of stimulation. What are the strategies for a human-centered AI? Some of the strategies for a human-centered AI include inclusive design, so in other words engaging a diverse group of people in development of AI systems, so we can have their opinions and their views reflected into the AI systems. Ethical AI governance that concerns developing and enforcing guidelines policies and regulations to guide ethical AI development and use. Continuous monitoring is also important to ensure any regulatory act is implemented so that is to regularly access AI systems for ethical compliance and societal impact.

25:30

All right, time to talk about machine learning. A bit earlier, we talked about a rational agent. And we mentioned that for a rational agent to act rationally, it needs to learn from the environment. agent to act rationally needs to learn from the environment. It

needs to get feedback from its actions and then incorporate those feedback into the future decision-making. That is machine learning. So the machine's ability to learn. That gives computers the ability to learn without being explicitly programmed. That means the computer is not aware of all possible scenarios but rather relies on the machine's ability or the algorithms that are built into the machine's ability to learn from their interactions with the environment to make rational decisions or human-like decisions about future situations or scenarios. We can see the relationship between artificial intelligence, machine learning and deep learning. Deep learning is a form of learning that is based on a kind of human-like learning or human brain-like learning or brain human brain like logical structure that is referred to as neural networks or artificial neural networks. So I've included a video here that summarizes some of the main aspects of machine learning in just a few minutes. You can watch this video by clicking on the video link embedded in the slides. All right, so the machine wants to learn from the environment and its data. The data could be labeled or not labeled. If the data is labeled, that means if the outcome is known for a set of inputs, then we're doing supervised learning. We can use this data that is labeled to train a classifier for instance, so they can tell us whether an email they sent is spam or not based on its characteristics and the labeled dataset that we have. What's the labeled dataset in this case? A set of spam emails. So we know that these kind of emails, that means the emails that has these certain characteristics are spam. We know that from the data that we have. Let's say someone has gone through them manually and has labeled them as spam, spam, spam, spam. This one is not spam. This one is spam. So they have been labeled as spam, not spam. Zero, one. So we can use that to train a classifier so that the classifier can build a model. The classifier, in fact, is the model. Or the computer can build a model that is the classifier to be able to tell whether the next email that's coming to your mailbox is spam or is not spam. You can classify the email. Another example is image classification. Given an image, classify it into one of several categories such as dog, cat, birds. So again, there could be a sensor there. It's a camera. there could be a sensor there, it's a camera, it captures photo of animal and then the classifier will be able to tell whether this animal is a cat or a dog. And that still needs a set of training data that is coming from a data set of images of different animals again labeled. Usually the initial data set is labeled manually for training. And that's the task of classification which is a type of supervised learning. So if you pay attention in examples that we export in the case of classification, the labels are categories like Z01 or animal 123. Sometimes the labels are continuous. When the labels are continuous we can

n use other forms of supervised learning regression models to be able to make predictions about the future values which are continuous. An example is to predict the house price based on let's say the number of bedrooms and the land size and number of bathrooms are the parameters, maybe the location. Another example is predicting stock price, given historical stock price. So these are some examples of regression which is a type of supervised learning. which is a type of supervised learning. So in both cases, in classification and regression, we need labeled training data sets. Usually the data set is split into like 70%, 30%, where 70% is used to train the classifier, 70% is used to train the classifier, and 30% is used to test the classifier in terms of the different parameters that exist, such as precision and recall and other measures that exist.

03:09

Okay, now let's talk about the alternative for supervised learning, unsupervised learning. Unsupervised learning is a form of machine learning where the data set is not labeled. We don't have a labeled data set. In the case of supervised learning, we could just look at the data and use the labels to train a classifier, for instance. But here we don't have that labeled data set. We just have the data, but the data is not labeled. There are two types of unsupervised learning that are commonly used. Clustering and dimensionality reduction. In the case of clustering, we try to form categories or groups from the data that we have. If you remember from the previous slide, supervised learning, in the case of clustering, for example, we had a data set of animal images. And we would want to tell whether it's a dog or cat or whatever other animals. Some of these data entities were labeled and we would use them as training data set for the classifier to do supervised learning. But now we don't have that data set. So what can we do? We can cluster animals into different groups, which may not necessarily be exclusive. They might have overlap. For example, we might have a cluster or a group that contains all of the birds but not all birds. And then you might have another cluster that might include a tiny proportion of the birds and most of the mammals, for example. So these clusters might overlap, may not be precise, may not be completely distinguishable. So it's kind of a try and error process. We use some parameters to form the clusters or groups. And then we look into them and try to make adjustments to make them as clear as possible. To then form groups. Then we can look at them and label them as okay the animals that are falling into this particular category are for example all dogs or the majority of them like 99 of dogs and there could be one percent of cats that might fall into this category but that's fine. That's the error that we

accept. And then you have another category that comprises maybe 99% of the cats and 1% of the dogs. Something like that. And then you form these groups, and then based on these groups, then you can train a classifier. You can use this for classifying. Another example of unsupervised learning is dimensionality reduction. That is where we have a set of features that determine whether animal is a cat or dog or whatever. But we don't really need all of these features. We just need the, what do you call it, the minimum of the features, which are the determining features. And the rest of the features are not that important. In fact, it could be a function of the main features or the most important features in some cases. So one of the techniques that is used for accomplishing dimensionality reduction or reducing the number of the features to the absolute essentials is or the most important is principal component analysis PCA which is widely used for example in problems of face recognition and many other problems. In the case of another example of dimensionality reduction concerns sentiment analysis, that is to extract the most important features from text and this area techniques such as SVD or singular value decomposition can be used. So these are the examples of dimensionality reduction. And we of course talked about clustering and examples of clustering that could be customer segmentation or image segmentation. you So what kind of technologies do we need for NLP? We need machine learning and deep learning models. Deep learning is part of machine learning. We need natural language toolkits such as for example NLTK or spacing Python and we need advanced algorithms such as transformers. So these are some technologies we need for NLP. Some of the applications of NLP, most technologies are used for NLP. Some of the applications of NLP includes chatbots for customer services, market research, machine translation services, speech recognition and virtual assistant. And of course the application goes beyond these areas and to many other interesting areas of application these days, especially with the advent of generative AI models or large language models, including ChatGPT. So there are some key challenges that natural language processing needs to address. One is ambiguity in language interpretation. That's still cancer. Another one is contextual nuances and cultural differences, which also affects the ambiguity of the language. So this relationship between these challenges. And still, sarcasm, irony and humor is hard for the existing NLP techniques to understand and interpret. Again, you can see the relationship between these three challenges. It's evident.

Okay, artificial intelligence, data science, machine learning, NLP, deep learning, these concepts are so interconnected these days and so merged into each other, it's really hard to tell them apart. So one might argue that data science and AI are no longer separate fields, but they are one. Not everybody of course shares this opinion depending how passionate they are about the independence of these fields but in reality you could see that they're highly interconnected. If you look at the applications such as chatGPT you can see that it's really hard to tell which part is data science and which part is AI. They are connected with dealing with data, we're dealing with text, we're dealing with natural language processing all combined together. But the relationship is we can safely say that there's a huge overlap between these two fields. So a lot of the things that you learn as data science are actually components of AI and vice versa. But what is the classic definition of data science? So we talked about the overlap and everything that we talked about, the interconnection between AI, machine learning, and data science. But what is the classic definition of data science? Data science is an emerging discipline. It remains a science where new knowledge and tools are still being invented. So, I'll say the same thing about AI. There's not yet a clear definition agreed by all for term data science. Different definitions exist from different perspectives, government, businesses, research, etc. Let's have a look at a definition of data science from NIST, National Institute of Standards and Technology. Their definition is as follows. Data science is the methodology for the synthesis of useful knowledge directly from data through a process of discovery or hypothesis formulation and testing. So the focus is on the methodology, on the steps, on the methods that are used for analyzing data through a process of discovery or hypothesis testing, formulation and testing. But what is the definition of data scientist based on NIST? A data scientist is a practitioner with sufficient knowledge in the overlapping regimes of business needs, domain knowledge, analytical skills and software and systems engineering to manage the end-to-end data processes in the analytics lifecycle. So I know this sounds confusing but I like this definition because it also includes me as a data scientist, even though my background is software engineering, but it says software and systems engineering. So, yes, I'm also included and I appreciate this inclusiveness. In fact, what it's trying to say is that a data scientist needs to have a good understanding of the fields that are related to analyzing data. That includes, of course, all the disciplines involved. And it's an inclusive definition, but also we need to have in mind that not everybody has all the knowledge in all these areas. So it requires team effort to efficiently analyze data. It needs people from different perspectives and of course those people

need to have some understanding of other areas. Like me as a software engineer or someone with background in software engineering, I need to have a reasonable understanding of AI and data analysis and other related fields if I want to be known as a data scientist or if I want to work as a data scientist. Alright so who is a data scientist? There's a bad joke from Joel Gross. I'm deliberately not pronouncing it as Joel Gross, even so I don't know which was correct. There's a joke that says a data scientist is someone who knows more statistics than a computer scientist and more computer science than a statistician. And he says, I didn't say it was a good joke. To be politically right. In fact, some data scientists are, for all practical purposes, statisticians, while others are fairly indistinguishable from software engineers. Again, I'm included. And some are machine learning experts, while others couldn't machine learn their way out of kindergarten. So, bad joke. So, data science is interdisciplinary. Everyone is included. AI experts, machine learning experts, statisticians, software engineers, and you name it. So one could say a data scientist must be a jack of all trades. But what is our focus? As we discussed earlier, we focus on the synergies between data science and AI. We look at data science and AI as one area, as one domain, with huge overlap. We focus on the synergies. We focus on analyzing textual data in this unit. So we talk about NLP, we talk about how we get access to textual data, how to clean textual data, how to analyze textual data, how to visualize textual data, and how to process natural language again in textual format.

46:50

Alright, now let's say we want to do data analysis. There are things that we need to do to prepare for that before we analyze the data. We need to understand what is our purpose, what is our goal of analyzing the data. So to set your goals helps you to have a vision, helps you to guide your decision-making throughout the process of data analysis. Helps you to have focus, to be able to measure the accuracy or the performance of your data analysis. You can track your progress over time, evaluate the impact of your analysis, and make informed decisions about next steps. Setting goals also helps you with collaboration. It helps you to align your stakeholders and collaborators around the shared region. It also helps you avoid confusion and misunderstanding. It also contributes to accountability. Setting goals helps you hold yourself and others involved in data analysis accountable for the results. Another important thing you need to do before starting data analysis is to define your audience. You need to know who you are doing the data analysis for. You need to know if your audience is going

to be a manager or a client, other researchers perhaps, and then you need to tailor your analysis and presentations, the type of visualization that you create for instance, to their specific needs. So if you're presenting your data analysis to a manager, they would not be interested in knowing the details of how to interact with your data and how to perhaps carry out more sophisticated analysis on the data. carry out more sophisticated analysis on the data. They just want to know some high-level insights and clearly communicated with them and perhaps the impact of the insights that are extracted from the data. So they're interested in high-level insights. If you're doing data analysis for clients, that could be different. They might be more interested to know about how your analysis can help them to solve specific problems or how to use specific tools for further exploring the data themselves and be able to perhaps create different reports from your data or for more sophisticated data analysis from the components that you provide to them so they might have such interests which you need to of course tailor your data analysis to their needs data analysis to their needs. So to tailor your data analysis to the audience it also helps you to expect or anticipate potential objections or questions and kind of predict that, foresee that and incorporate that into analysis so that your credibility and trust is increased in the eyes of the target audience. Once you've specified your goal and your target audience, it's time to work on the data. And the first step or the first phase of data analysis is to get the data. You can get the data or, in other words, collect the data, acquire the data from different sources. You can collect the data from an existing database. Or you can scrape the web. Or you can use some experiments to code data. You may also use APIs. API stands for Application Programming Interface. It works like a connection point. In a way, it resembles a power plug. a power plug. So an API gives you access to another software or to another or to a data set where you can get the data from that data set or from that software through the API. So API is an interface in a way that gives you access to a data set or to a software that has access to that data set. For instance, you can use a Twitter API or X API. You can use a YouTube API or other types of APIs that exist. You can use a GitHub API. You can use a YouTube API or other types of APIs that exist. You can use a github API to get access to different repositories and collect data from them and then use them for your analysis. You also need to document the data that you're collecting, the sources of your data, and other information about your data, including privacy matters. That is to help you with governing your data, making your analysis reproducible, so that if other people want to repeat your experiments or you repeat your analysis, they can repeat it and they can get the same results u

using the same data set. And of course, it assists with transparency. So documenting the data that you collect, the data source, and the process of collecting the data is an important component of getting the data.

53:11

The second phase is to clean the data. So once you've collected the data, the data may not come in a form that is ideal for your type of analysis. That includes removing unnecessary rows or columns in your data. We're talking about textual data that is structured in terms of rows and columns, the table. So you may not need some of the entities in the table. That means the rows, or you may not need some of the attributes of these entities, which are the columns. So you can get rid of those to make your data more appropriate for your analysis, which will reduce confusion, prevent from error, and also increases the speed of analysis. Your data may also include missing values. So you might have missing values for certain entities. You need to make a decision as to how you want to treat them. There are different approaches to this. You can get rid of entities or roles which include missing data. get rid of entities or roles which include missing data or you can try to use techniques such as substitution or imputation to sort of to sort of predict or estimate the values which are not present based on the relationship between or statistical relationships between other values. Sometimes you have the data and there's no missing data, but the data is not coming or the values are not coming in the format that are appropriate for your analysis. For example you have datetime values which are coming in terms of which are formatted as text or objects. Whereas you want to analyze them as datetime objects. So you might need to convert them to a format that makes it easier to analyze them. That is also part of data cleaning. So data cleaning involves removing unnecessary rows and columns, that means data entities and their attributes when they're not relevant to our analysis, handling invalid or missing values, and changing object data types to, for example, daytime or numeric types. So the third phase that I want to talk about is data preparation or prepare the data. That involves adding new columns sometimes, which contain values that are calculated based on other columns. That makes the analysis easier. So that you don't have to, for example, make these calculations when you're preparing a visualization. You can perform these calculations before any further analysis, and then use the values in your more sophisticated analysis. You may also want to restructure the data in a format that makes it easier to work with. That includes you might want to filter the data, sort the data or group data or group your data entities

based on different parameters or based on different column values. You may also need to make or prepare some preliminary visualizations for better understanding the data and one of the things that you can understand is outliers. You can look at the data visualization and see if certain data entities do not match the rest of the data in terms of the values or characteristics and then you could further investigate them. This is a basic visualization that you would do at the data preparation phase before proceeding to more sophisticated analysis.

57:37

All right, data analysis phase four. In phase four, we analyze the data. So once we collect the data, clean the data, prepare the data, it's now time to analyze the data. To do that, we might be interested to get new views of the data by grouping and aggregating the data. We group the data to summarize the data and then we apply aggregation functions such as sum, average, count to calculate summary statistics for each group. We may also create visualizations to provide insights and show relationships. So at this stage we use more sophisticated visualization techniques. This involves creating charts, graphs and other visual representations. There are all different types of graphs that you can create such as heat maps, pie charts, sunburn charts, you name it. Tree like visualizations. So there are many different ways of visualizing data, which might be appropriate for different purposes, depending on the details that you want to show and the relationships you want to show. So you might want to try different types of visualizations to eventually find the ones that is appropriate for your goal and your audience. appropriate for your goal and your audience. You may also be interested to model the data as part of a predictive model. So you may want to create a predictive model or do predictive analysis and for that you might build statistical models such as regression or decision trees to predict future outcomes based on the data that is in your data set, the data that you have collected. So you might want to make predictions to identify the key factors that drive certain outcomes. Phase five and phase four are highly connected. So it's very hard to tell them apart, but here we have separated them so we can further emphasize on visualization. The visualization starts from very early stages from where we actually prepare the data. So we prepare some preliminary visualization in the data preparation phase. And as we continue to the analysis, we create more advanced visualizations. And then in phase five, we enhance the visualizations to make them appropriate for the target audience. And that involves tailoring the presentation of data We enhance the visualizations to make them app

ropriate for the target audience. That involves tailoring the presentation of data to effectively communicate insights and key messages to a specific group of people. This can include modifying the design, layout, and level of detail of visualizations to match the audience's level of expertise, interests, and preferences. So we might have explored several different visualizations in the analysis phase, phase 4. In phase 5 we reduce those visualizations to what is most important for our target audience. So here we kind of improve and reduce our visualizations to what is required for our target audience. An example is a simple and intuitive visualization may be more suitable for a non-technical audience, while a more complex and detailed visualization may be more appropriate for a group of data experts. visualization may be more appropriate for a group of data experts. So here we have a question for you. What's the difference between data cleaning and data preparation? I want you to think about this for a moment. So the answer data cleaning focuses on identifying and correcting errors and inconsistencies in the data, while data preparation focuses on transforming and formatting the data to make it suitable for analysis. Both steps are important in the data analysis process as they help to ensure that the results of the analysis are accurate and meaningful.

01:02:09

Okay, Python for data science and AI. I'm sure you all know that Python is the primary programming language that is used nowadays for analyzing data science and AI tasks. If you've used ChatGPT, you've seen that ChatGPT uses Python to communicate solutions for data science and AI tasks. So we need to learn Python so we can formulate a problem, a data science or AI problem, write down a solution for it and execute the solution. We need that. In this unit we're going to use a tool called JupyterLab that comes as part of Anaconda. Anaconda is another tool that gives you a combination of or a variety of tools that are used for programming in Python. So there's a whole different set of IDEs or integrated development environments and other tools that can be used as part of the Anaconda. So you're going to install Anaconda in your first workshop and learn how to work with it. The tool that we are using as part of Anaconda, which you could install independently, but it comes with Anaconda, is JupyterLab. We use JupyterLab for writing and executing Python code as the default tool in this unit. However, you're welcome to explore other tools and you have preference for other tools or you want to use them, feel free to do so. But we use JupyterLab as the standard tool in this unit. So with Anaconda, there are some modules that are inclu

ded. They include Pandas, NumPy, Seaborn. Seaborn is used for visualization. Pandas allows you to work with the data, allows you to transform your tabular dataset into a data frame and then work with the data frame. We'll talk about these in detail in the coming weeks. And another module. So in Python, we use modules to benefit from certain functionalities, to utilize certain functionalities for carrying out data science and AI tasks. So when we want to, for example, work with some of the... For example, we want to work with datetime objects, we should use the datetime module. So we need to import this module, datetime, so we can work with it. If you want to do some visualization using seaborn, you should import seaborn and then work with the functions that are imported, that come as part of seaborn. We always need some reference to these modules to work with the functions that come as part of these modules. For example, we can import pandas as pd, numpy as np. These abbreviations could be anything you could just have your own but there are some sort of standard abbreviations that you can stick with. Usually pandas reference is named pd, numpy is named np, seaborn reference is abbreviated as sns and so on so forth. But again you can just use your own convention and that's fine. So there are modules and sometimes within the modules there are sub modules that can be imported. A lot of the modules that you use in this unit come as part of your default installation of Anaconda but if you need to install a specific module you can do so. The way Anaconda organizes the modules is that it organizes them under different channels. If you use conda install pandas command in your shell environment, that will install the pandas module for you from the default channel of Anaconda. If you want to install a module from a specific channel, you can use the -c channel parameter. So conda install -c conda-forge pandas. Conda-forge is the name of the channel and then the name of the module that comes after that. You can also include multiple channels. So using the same -c channel module name syntax. What else? So CondaForge is community driven channel for Conda package that provides many additional packages beyond those available in the default Conda channels. And of course you can view the documentation of an account for different channels and how we can search in them and how we can install in them and how you can install different modules from them. The link is provided here.

01:08:15

Assuming you have installed the modules you need, you may want to import them in your program and use them. And obviously you need the import statement. Import

pandas is the name of the module and here you use as keyword again, PD. PD is a reference by which you can refer to the functionalities or the functions that come with pandas. PD is just a reference. You can name it however you like. But the way it's named here, PD, is the convention. You may also want to import a module that is embedded within a larger module. So in this case, we have URLLib, and URLLib contains another module that is request. So request is inside URLLib. You would want to import from URLLib a sub-module that is named request. In this case, you don't need a reference. So you can just directly refer to request and use it in your code. Alternatively you could have imported urllib and every time you wanted to use a function from request you could say urllib.request. the name of the function and use it. This is the easier way to directly use functions that come from requests and you're also specifying down I only need these modules, this sub module from URLeap so any other sub modules that are in URLeap are not going to be used in this Python script which makes it more efficient in terms of interpretation of the script. Alright, so once you imported a module you want to get access to the functions of that module and use them. But before we talk about that, you see a keyword here, method. How to call a method? Why are we talking about methods here? Method and function are used interchangeably in this unit, but they're not exactly the same. Method is a concept that comes from object-oriented programming. Python supports object-oriented programming, but it's not a pure object-oriented programming language, as opposed to languages such as Java. Java is a pure object-oriented programming language, is a pure object-oriented programming language, or used to be, or was designed with that mentality. Whereas Python was a procedural language, which later on adopted some of the concepts from object-oriented programming and supported object-oriented programming. So that's why the term method could be used to refer to the same concept of functions. Method corresponds to type of functionality that is a functionality that comes as part of the behavior in object not in isolation. So in other words in object oriented programming every behavior, every function must be attributed to an entity or an object and when that can be done the function is no longer called function, it's called method. It's a method of a specific object. But we leave that discussion out for now and we just clarify that the term method and function for the purpose of this unit are the same. We use them interchangeably. So how to call a method or function in a module and import pandas as pd? First we import of course the module and then use pd which is a reference to the module pandas and then use dot and then we use the name of the function that I want to use which is read CSV in this case. Re

ad CSV is going to read a CSV file for us in this case from the URL that's provided here. This URL refers to a CSV file and creates a data frame. Data frame is a type or a type of object so what read CSV does it reads the CSV file and then creates an internal object for us that we can interact with we can work with in our Python code and that object is a data frame. We refer to that data frame as `Polls`. So `P-O-U-L-E-S`. This is a reference to the object that has been created by the function `read CSV` and returned to us. Okay. Return to our code. We refer to that object as `polls`. This could be named anything. It's just a variable name, it's a reference. And then we can use `polls` to use other functions that are associated with the data frame. For example we can sort values. So `polls.sort_values` and sort values based on start date or based on any other attribute of the, or any other column of the data frame. So, the columns of the data frame represent its attributes and we can use them to sort our data frame accordingly.

01:14:35

Method chaining, how to chain methods. In Python we can call a method or a function and then provide the output of that function as the input of the next function. This is called method chaining and we can keep doing that. So here's an example. `Polls` which is a data frame or yeah is a data frame does `sort_values`. `Sort values` is the function that we are calling. It's going to sort the values or it's going to sort the rows of `polls` based on start date in ascending order by default. We can specify otherwise if we want to but by default it sorts in ascending order based on start date. So the sorted data frame, the data frame that is sorted based on the start date in ascending order, will be provided as the input of the next function that is `head`. The `head` function is going to take the first five rows of the sorted data frame to us. So this is what this method chain does. And of course you can print the results, which would be another function that you can call. The next example here uses a `query` function. So the `query` function is going to return a subset of the data frame based on the `query` function. So the `query` function is going to return a subset of the data frame based on the `query` parameter. In this case you're specifying that I want all the records from the data frame except the ones whose state value is US. In other words, exclude any record, any row, whose state value is US. And then the next function, the `plot` function, takes this data frame as its input and plots the data plots what we are plotting here, the x and y, the x parameter specifies the start date. So it plots the election results for two candidates, Clinton and Trump, whatever the PCT represents against the start date. So the x-axis is the start date and the y-axis represents the election results for different two cand

idates. How to call a method with parameters? So the methods have usually parameters and you need to specify the values for those parameters when you call the method or the function. And these parameters sometimes are positional, so you pass the values that need to be assigned to the parameters based on their positions in the sequence of the arguments, in the sequence of the parameters or directly specify the values so here's an example for example start date here is the first parameter that is passed so we don't need to know what's the what's the what you call it the keyword for start date in that in the target function, which is calling the function, sort values and that keyword, in other words the parameter name is y. By is used in the function, in the definition of functions, function or in the signature of the function to work with the start date. But when we're calling the function we just provide with the start date. But when we're calling the start date function, we just provide the value, start date. This is based on the position. Or we can specify directly using the keyword or using the name of the parameter. You can see here, ascending is true. So we're setting ascending to, this is the default value, ascending true. We're passing ascending false., so we're setting ascending to, this is the default value, ascending true, we're passing ascending false. If we don't pass any value, the value true will be used by default. So this sort function is going to sort in ascending order, unless we specify that we want to sort in descending order, in that case we'll assign the value of false to the parameter ascending. And then there's another parameter in place that specifies whether the sort needs to be, the sort operation needs to be performed on the actual data frame or on a copy of the data frame. If it's true, it would be performed on the actual data frame. If it's false, it will be performed on a copy of the data frame. And then the copy will be returned by the function. So, yeah, so we can, there are two different ways that we can pass the, specify the parameters. we can pass the specified parameters. List, slice, tuple and dictionary. In Python, you can create a list of items, comma separated items, or you can create a tuple. And the only difference into the syntax here is using bracket as opposed to parenthesis. But in terms of how they work, there's a significant difference between list and tuple. And that is tuples are not mutable, which means once the values are assigned to a tuple, they cannot change during the operations. We don't have this limitation for the lists. So in a way it's limitation, but it depends how you want to use them. In fact, in some cases, it's preferred to use the tuples because you wanna make sure that the values cannot be changed for consistency and to prevent from error. Also, we have dictionaries. Dictionaries in dictionary which item is not a singular item, is a pair. It's a pair of key and value. We want to have a key by w

hich we can refer to the values. For example, let's say you use student ID to refer to the student to extract their corresponding grades or to extract their corresponding contact numbers or emails or any other information they want to use. So keys are used in a way that they can uniquely identify the entities or the items in the dictionary, but the values are not necessarily unique. Take, for example, the example that I just gave you. Student grades are not unique. That means two students can have the same grade, but the student ID is unique. So, but the student ID is unique so keys should be unique whereas values are not necessarily unique. We can also get a slice of a list, we can have start, stop, step. You can get a subset of a list.

01:21:56

Alright so here we have some examples of using list, slice, tuple and dictionary. We'll have more examples in workshop week one and you can have more practice there. The first example here shows the use of list. So we have passed a list of column names as the value for these parameter columns of function drop. So we have applied this function drop to the data frame pulse to drop these columns from the data frame. So the resulting data frame does not contain the values corresponding to these columns. These columns are absent in the resulting data frame. Since the parameter inplace is set to true, this is going to affect the original data frame pulse, not a copy of the data frame but the actual data frame, the original data frame. The next example shows passing a tuple with two values, perhaps start date and end date, which specify the range of values that are going to be displayed on the x-axis of the line plot. So the xlim is set to this is the start, this is the end and this creates a line plot for this range. The third example is a dictionary, shows the use of a dictionary, and we're using that to rename the columns of a dataframe, dataframe polls. So we're renaming the columns of the dataframe as follows. The keys in the dictionary specify the existing names of the columns or the existing labels and the values specify the new names or new columns or new labels. So this name here will be changed to Clinton and this name here will be changed to Trump. So the resulting data frame will have different names for these columns. This is how the rename takes place. Finally we have an example of using slices. So we have used this slice as the input of or as part of the loc accessor. So the loc accessor will locate rows and columns of the data frame as follows. We're specifying a slice here, 0 to 100, and the step is 10. So from row 0 to row 100, we are picking by the step of 10. So we're picking the rows, not all rows, but we're picking the rows with the step of 10 starting from 0 up to 100 and the second slide shows the

second slide shows that we are only interested in the columns between state and grade. So it includes the column named state and the column named grade. It includes every column between these two columns, including these two columns, and excludes any other columns. So any column that comes before state and any column that comes after grade in the sequence of the columns will be excluded. So this function eventually, or this accessor eventually, will return a subset of the data set with these specific columns between state and grade. So it gives you a horizontal and a vertical slice of the data frame. In Python, you can also use list comprehensions. Here's an example of a list comprehension. least comprehensions. Here's an example of a least comprehension. We're extracting x's for x in a range of 900 to 920 with a step of 2. So the values that are extracted are as follows 900, 902, 904 all the way up to 918. So these are the values that are extracted for the for example for the x for the labels on the x-axis of a for example plot. So we want to see So we want to see values that correspond to these values in the plot. And these values start from 900 go all the way up to 918 with the step of 2. So the least comprehension here is that we are performing an operation, we are iterating over a range and extracting certain values and returning the result as a list. The type function. Sometimes we have an object in our program and we're not sure what type it is, especially when you're calling a function and a function is not known to us, it returns certain values and we don't know what type it is, especially when you're calling a function and a function is not known to us, it returns certain values and we don't know what's the type of the value. This is important to know how we can work with the returned value. So to understand the type of the return value so we can work with it, we can use this function type here you can see an example type poll url it's returning str which shows this variable poll url is a string or for example type of polls is a data frame so pandas that for that frame that data frame this shows like the hierarchy of the object shows data frame is part of frame, frame is part of core, core is part of pandas and this pose is a data frame or is an object of type data frame and sometimes the statements you're writing are too long and they do not fit in one line and you would prefer to break them into multiple lines and you know that in Python indentation is important so that indentation is meaningful specifies the beginning and end of the blocks so you need to do this properly and you need to break this one lengthy line into multiple lines properly so it doesn't lead to syntax error. by just hitting the enter. And sometimes you need to use the backslash symbol to specify this is the end of the line. And this is going to be followed by the next line of the statement. Two ways that you can break instructions into multiple li

nes.

01:29:45

As I mentioned earlier, we use JupyterLab in this unit as the standard tool for writing and executing Python code. So get yourself familiar with it. So counts as part of the Anaconda package and it's straightforward to use. In workshop week one, you're going to learn how to work with JupyterLab and how to write and execute Python code in the code cells, how to navigate around JupyterLab. around JupyterLab. So there's a video here that gives you a quick introduction to JupyterLab and there's another video here that contains a tutorial on JupyterLab. It's also included in your workshop week one contents. One thing that you want to pay attention to is that it might be a bit tricky to change your working directory in especially working on Windows for JupyterLab. So the easiest way to easily upload your workshop contents and start working with them is to use the upload option here so just use just upload your files for using this you can upload all the all the contents of your workshop and start working with them here that's probably the easiest way to get started and of course then you can learn how to work with more sophisticated features. If you're on a Mac it's pretty straightforward you can just navigate to the directory that you want to execute JupyterLab from in your shell and you can just type JupyterLab and that will launch JupyterLab for you from that particular directory so you can easily get access to your directory from shell. In Windows it might be a bit more tricky. I'm not quite sure about that so this solution probably works best for everyone regardless of the operating system the y're using just directly upload your files and start working with your uh your python uh with your jupyter notebooks in jupyter lab environment using this upload button here so there are two types of error that may occur in your Python scripts. One is syntax error and the other one is runtime error. So syntax error mainly concerns errors that come from for example missing parentheses or misspelling a keyword things like that. Things that concern the syntax of your code. And they're usually picked up by the syntax trigger before the execution. And you can fix them before your script is, or your Python code is executed. In some cases you can, if it's too complicated or if the information you get about the syntax error is not clear, you can put your code to chat GPT and it will give you more detailed information. In most cases it's more informative than what you could get in your execution environment. Runtime error is another type of error that you can get that mainly concerns exceptions. For example you try to do some illegal operation. For example you want to do you mistakenly try to divid

divide a number by zero. Obviously you're not gonna type x divided by 0 but the value that you are using as the denominator might happen to contain the value of 0 and that could lead to a runtime error. So you should be aware of such errors and try to predict them and prevent from them. One efficient way to prevent from or in other words to handle such error is to use exception handling. So you can put a block that is risky and there's a possibility of error in it in a try block and then catch the exceptions when they happen but that's beyond what we discussed here just I just want you to be aware that there are two types of error syntax and runtime error and what else is important here and what else is important here? Yeah, so the runtime errors could break your program during the execution, whereas the syntax error, with the syntax errors, your program will not proceed to the execution. So those are the type of error that would be picked up by the syntax checker before you can even execute your code. But runtimes will occur during the execution of your code, so in fact they are more tricky to catch them and to fix them. Alright this will wrap up our lecture week 1 part 2 and see you in the next lecture.